

5635133

B.Tech. DEGREE EXAMINATION, SEPTEMBER 2021.

Fifth Semester

Computer Science Engineering

LANGUAGE TRANSLATORS

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. What is the difference between system software and application software?
2. Describe the purpose of pass 1 and pass 2 of a 2-pass assembler.
3. Define Absolute loader.
4. Brief on OPTAB and SYMTAB.
5. List out the objectives of compiler optimizations.
6. Define the role of lexical analyzer.
7. State the use of context free grammar.
8. Mention the applications of back patching.

9. List out the issues in the design of code generator.
10. Give the criteria for code improving transformations.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, one from each Unit.

UNIT I

11. Explain the various addressing modes and instruction formats of SIC/XC machine.

Or

12. Define the basic functions of assemblers and explain in detail about the two-pass assembler algorithm.

UNIT II

13. Why linking loader requires 2 passes to complete loading? Explain the functions of linking loader.

Or

14. (a) How can you load and call subroutines using dynamic linking? (6)
- (b) Explain the difference between linking loader and linkage editor. (5)

UNIT III

15. (a) Explain the different phases of a compiler. (5)
- (b) Write short notes on compiler construction tools. (6)

Or

16. Explain in detail about recognition of tokens.

UNIT IV

17. Explain briefly SLR parsing algorithm.

Or

18. How declarations are translated into intermediate code.

UNIT V

19. Discuss in detail about :
- (a) Copy propagation. (5)
- (b) Common subexpression elimination. (6)

Or

20. (a) Explain briefly about the design of code generator. (5)
- (b) Write the procedure for generating code from DAGS. (6)

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SEPTEMBER 2020.

Fifth Semester

Computer Science and Engineering

LANGUAGE TRANSLATORS

(2013-14 Regulations)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

1. Compare Interpreter and Assembler.
2. Define assembler directives.
3. What is loader? Mention the advantages and disadvantages.
4. Define Bootstrap loader.
5. State the role of Lexical Analyzer.
6. Define Cousins of Compiler.
7. What do you mean by back patching?

8. Compare predictive parser and LR parser.

9. Define Peep hole optimization.

10. State any two issues in the design of code generator.

PART B — (5 × 11 = 55 marks)

11. Discuss different design options of assembler. (11)

Or

12. (a) Discuss the steps in Pass 1 of two pass algorithm.

(b) Explain data structures used in assembler algorithm. (6+5)

13. What is dynamic loading? List out the advantages and disadvantages. Explain with a neat diagram loading and calling of subroutine using dynamic linking. (11)

Or

14. Explain in brief :

(5 + 6)

(a) MS-DOS linker

(b) CRAY-MPP linker.

15. List and explain the various phases of Compiler and show the output of each phase for expression $a = b + c * 25$.

Or

16. Name any three Compiler construction techniques and explain each.

17. What is Shift Reduce Parser? Explain the errors that may occur during shift reduce parsing.

Or

18. Write a note on Intermediate Code Generation. Generate Intermediate Code that finds Factorial of a number.

19. (a) Construct a DAG for the expression (6)

$$a + b * (a + b) + c + d$$

(b) Write a note on common sub expression elimination.

Or

20. (a) Briefly explain the main issues in code generation.

(b) Write a note on dead code elimination.

- (a) Write detailed notes on Basic blocks and flow graphs. (6)
- (b) How would you construct a DAG for a Basic block? Explain with an example. (5)
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B.Tech. DEGREE EXAMINATION, JANUARY 2022.

Fifth Semester

Computer Science Engineering

LANGUAGE TRANSLATORS

(2013-14 Regulations)

Time : Three hours

Maximum : 75 marks

PART A – (10 × 2 = 20 marks)

Answer ALL the questions

1. Define the term system software
2. Differentiate between compiler and interpreter.
3. What is loader?
4. List the advantages of dynamic linking.
5. What are called tokens?
6. List the Cousins of Compiler.
7. State the various conflicts that occur during shift reduce parsing?

8. What is meant by Context free Grammar.
9. What is copy propagation in compiler design?
10. Give the applications of DAG

PART B – (5 × 11 = 55 marks)

Answer ALL questions.

UNIT I

11. What is an Operating System? Discuss various classifications of operating systems.

Or

12. (a) Draw the detailed flowchart of 2-pass assembler. (6)
- (b) Explain databased used in PASS-I assembler. (5)

UNIT II

13. Explain the Design of Dynamic Linking Loader along with example.

Or

14. Explain in detail about basic loader functions

UNIT III

15. (a) Explain briefly about compiler Construction Tools.
- (b) Explain in detail about Cousins of compiler Tools.

Or

16. (a) Differentiate between lexeme, token and pattern.
- (b) What are the issues in lexical analysis?

UNIT IV

17. Construct predictive parsing table for grammar $E \rightarrow E+E$ $E \rightarrow E^*E$ $E \rightarrow (E)E$ and parse the string $id+id*id$.

Or

18. Explain the Concepts of Back Patching.

UNIT V

19. (a) Explain induction variable elimination examples.
- (b) Write short notes on Loop Optimization.

Or

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B.Tech. DEGREE EXAMINATION, DECEMBER 2023.

Fifth Semester

Computer Science and Engineering

LANGUAGE TRANSLATOR

(2013-14 Regulations)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. What is the purpose of the following assembler directives (a) BYTE (b) WORD.
2. What is control section in a program block.
3. Explain the feature of automatic library search.
4. Define Loading, Relocation and Linking.
5. What is a Lexeme?
6. What is the purpose of a symbol table?
7. Define Left Factoring.

8. Write the steps to generate intermediate code for procedures.
9. Write any two purposes of registers in a simple code generation.
10. What is an Induction variable?

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

UNIT I

11. Explain in detail the different types of Instruction set in Machine Instructions.

Or

12. List any Ten differences between the features of Machine-dependent and Machine-independent Assembler.

UNIT II

13. What is a Linking Editor? Explain in detail how it is different from Linking Loaders.

Or

14. Describe in detail the design of an Absolute Loader using an Object program.

UNIT III

15. Explain in detail the different Phases of a Compiler.

Or

16. What is a Transition Diagram? Explain its purpose in lexical analysis.

UNIT IV

17. Explain the working of LR – Parsing Algorithm with an example.

Or

18. Draw the Directed Acyclic Graph (DAG) for the following expression.

$a = b * c$

$d = b$

$e = d * c$

$b = e$

$f = b + c$

$g = f + d.$

UNIT V

19. Explain the different types of Instructions and addressing modes of a Simple Target Machine Model.

Or

20. Describe in detail the steps involved in evaluating a constant expression.

UNIT V

Discuss the principle sources of code optimization in detail. (11)

Or

- (a) Explain in the DAG representation of the basic block with example
- (b) How to generate a code for a basic block from its dag representation? (6 + 5)

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B.Tech. DEGREE EXAMINATION, MARCH 2021.

Fifth Semester

Computer Science and Engineering

LANGUAGE TRANSLATORS

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Give some applications of OS.
2. What is meant by assembler directives. Give example.
3. What are the basic functions of loaders?
4. Give the difference between linking loader and linkage editors. Into the memory.
5. Mention the two parts of a compilation.
6. What is Lexical Analysis?
7. Mention the basic issues in parsing.

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UNIT III

8. List the various types of intermediate code representation.
9. List the properties of optimizing compiler.
10. Define DAG. Mention its applications.

PART B — (5 × 11 = 55 marks)

Answer ONE question from each Unit.

UNIT I

11. (a) Discuss about Instruction sets in SIC and SIC/XE.
- (b) Explain about the SIC architecture. (6 + 5)

Or

12. (a) Demonstrate in detail about basic assembler functions.
- (b) Explain in detail about the assembler Design options. (6 + 5)

UNIT II

13. Discuss about Machine-Dependent Loader Features. (11)

Or

14. Explain about the Loader Design Options. (11)

15. Briefly outline the concepts of phases of compilation with neat sketch.

Or

16. Discuss in detail about the role of Lexical syntax analyzer with the possible error actions.

UNIT IV

17. Illustrate the predictive parser for the grammar.

$S \rightarrow (L) | a$

$L \rightarrow L, S | S$

Analyze, Is it possible, by modifying the grammar in any way to construct predictive parser for language of $S \rightarrow SS^+ | SS^* | a$ the string "a"?

Or

18. (a) Explain about the concepts of Block Expressions.
- (b) Narrate the steps of procedure calls.